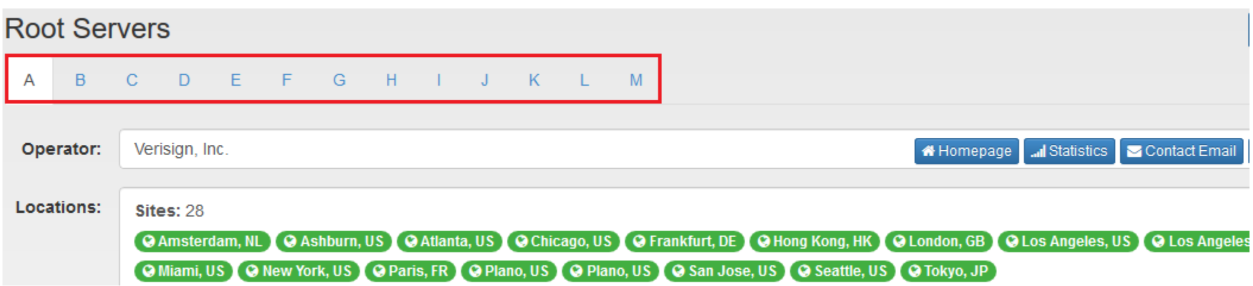
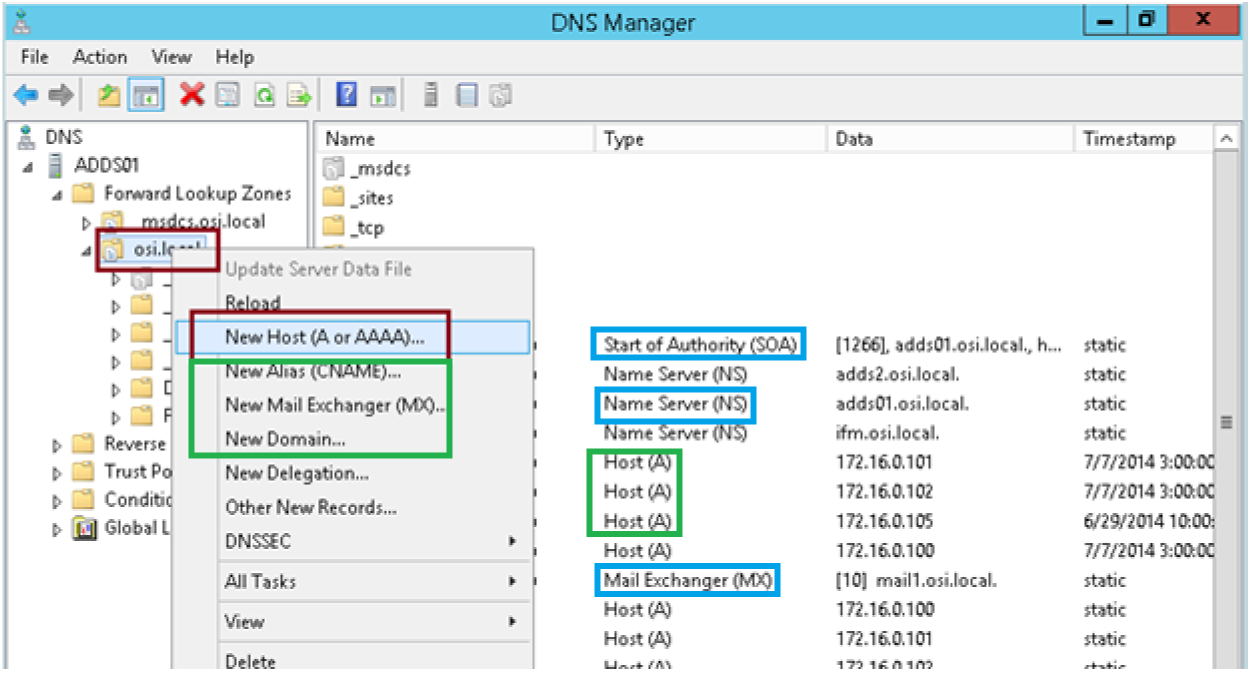
DNS Server:

- DNS Stands for Domain Name System or Domain Name Server.
- DNS is a large database, which resides on various computers.
- DNS contains names & IP addresses of hosts on the Internet & various domains.
- DNS servers match domain names to their associated IP addresses.
- The Domain Name Systems (DNS) is the phonebook of the Internet.
- DNS convert IP Address to domain name & domain name into IP address.
- DNS names are assigned through Internet Registries by the IANA.
- There are 13 root name servers from a.root-server.net to m.root-server.net.
- 13 DNS root name servers can be check on this link http://www.root-servers.org.
- DNS primarily uses User Datagram Protocol on port number 53 to serve requests.



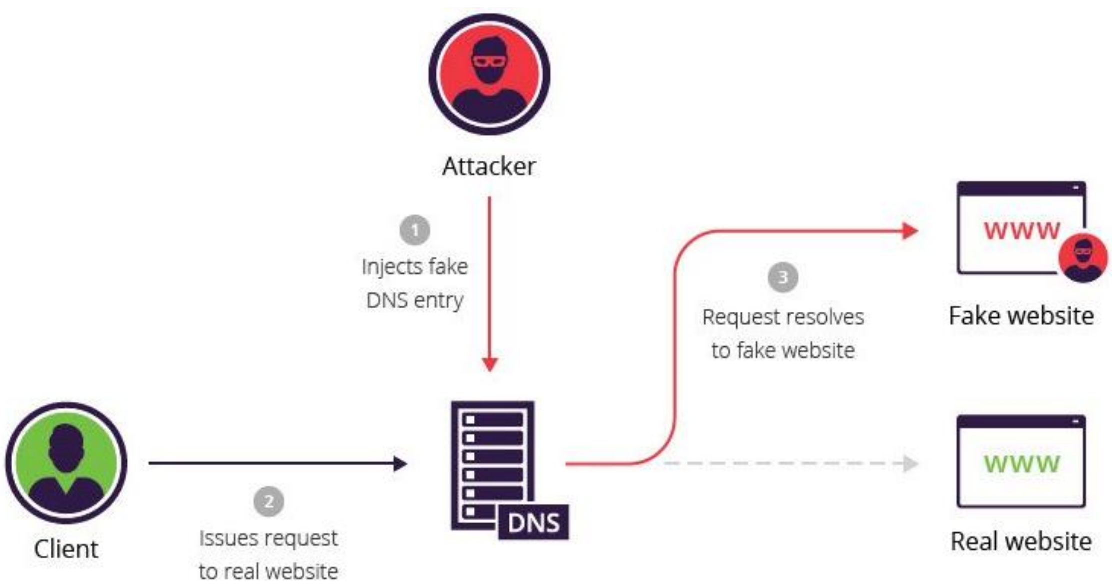
DNS Records:

- There are several different types of resource records used by DNS.
- The A record specifies Internet Protocol IP address (IPv4) for given host.
- A, records are used for conversion of domain names to correspond IP addresses.
- The AAAA record specifies Internet Protocol IPv6 address for given host.
- Domain name system also allows us to name single device but give it multiple names.
- Give it nickname or secondary name it has called a Canonical Name record, or CNAME.
- CNAME records are used for creating aliases of domain names in DNS Server.
- CNAME records are truly useful when we want to alias our domain to an external domain.
- The MX resource record specifies a Mail Exchange server for a DNS domain name.
- PTR stand for Pointer Record; this is opposite of an address record (A or AAAA).
- An address record took a name and provided you with an Internet Protocol IP address.
- A Pointer record took Internet Protocol IP address and come up with a name.
- Name Server (NS) The NS record specifies who the DNS servers are for the zone.
- Start of Authority (SOA) The SOA record stores the settings for the DNS zone.



DNS Spoofing:

DNS cache poisoning, also known as DNS spoofing, is a type of attack that exploits vulnerabilities in the domain name system (DNS) to divert Internet traffic away from legitimate servers and towards fake ones. DNS Poisoning is a technique that tricks a DNS server into believing that it has received authentic information when, in reality, it has not. For instance, a user types www.google.com, but the user is sent to another fraud site instead of being directed to Google's servers. As we understand, DNS poisoning is used to redirect the users to fake pages which are managed by the attackers.



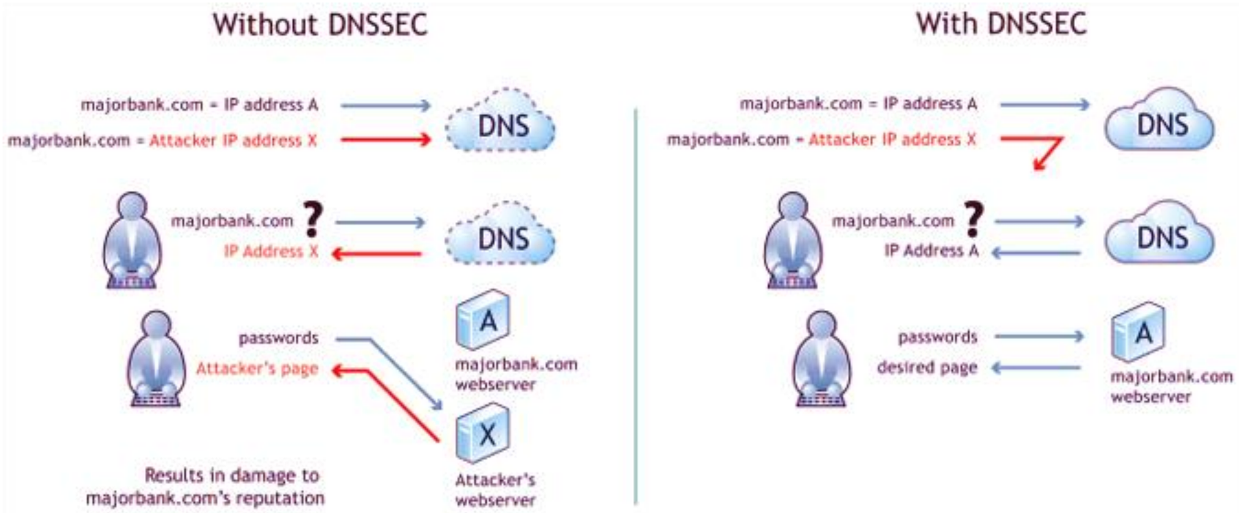
DNS Caching:

The Internet doesn't just have a single DNS server, as that would be extremely inefficient. Your Internet service provider runs its own DNS servers, which cache information from other DNS servers. Your home router functions as a DNS server, which caches information from your ISP's DNS servers. Your computer has a local DNS cache, so it can quickly refer to DNS lookups it's already performed rather than performing a DNS lookup over and over again.

```
C:\Users\aali.c>ipconfig /displaydns

Windows IP Configuration

    gem.gbc.criteo.com
    ----------------------------------------
    Record Name . . . . . : gem.gbc.criteo.com
    Record Type . . . . . : 5
    Time To Live  . . . . : 18403
    Data Length . . . . . : 8
    Section . . . . . . . : Answer
    CNAME Record  . . . . : gbc0.nl.eu.criteo.com


    Record Name . . . . . : gbc0.nl.eu.criteo.com
    Record Type . . . . . : 1
    Time To Live  . . . . : 18403
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 178.250.6.19
```

DNSSEC:

- DNSSEC is short for Domain Name System Security Extensions.
- It is a set of extensions that add extra security to the DNS protocol.
- This is done by enabling the validation of DNS requests.
- It is specifically effective against DNS spoofing attacks.
- DNSSEC provides the DNS records with a digital signature.
- Therefore, the resolver can check if the content is authentic.
- DNSSEC services protect against most of the threats to the DNS.
- DNSSEC mechanisms require changes to the DNS protocol.
- DNSSEC adds four new resource record types:
- Resource Record Signature (RRSIG), DNS Public Key (DNSKEY)
- Delegation Signer (DS) and last Next Secure (NSEC).
- DNSSEC feature helps to protect DNS traffic from threats.
- Security extensions adds chain of digital signatures in DNS hierarchy.
- Each level in the DNS hierarchy owns its own signature generating keys.
- Domain Name System clients get only authentic DNS responses.



Defenses Against DNS Poisoning:

- Implement IP DHCP Snooping on switches to prevent ARP poisoning & spoofing attacks.
- Implement policies to prevent promiscuous mode on network adapters or Lan Cards.
- When deploying wireless access points all traffic on wireless network is subject to sniffing.
- Encrypt your sensitive traffic using an encrypting protocol such as SSH or IPsec.
- Port security is used by switches that have ability to allow only specific MAC addresses.
- IPv6 has security benefits and options that Internet Protocol IPv4 does not have.
- Replacing protocols such as FTP & Telnet with SSH is an effective defense against sniffing.
- If SSH is not a viable solution, consider protecting older legacy protocols with IPsec.
- VPNs can provide an effective defense against sniffing due to their encryption aspect.
- Secure Sockets Layer (SSL) is a great defense along with Internet Protocol Security IPsec.

Make Router DNS Server
R1(config)# ip dns server R1(config)# ip domain-lookup R1(config)# ip name-server 8.8.8.8 R1(config)# ip domain name test.local R1(config)# ip host pc1.test.local 192.168.1.2 R1(config)# ip host fsrv.test.local 192.168.1.3 R1(config)# ip host pc2 192.168.1.4 R1(config)# do show hosts
Client
R2(config)# ip domain-lookup R2(config)# ip name-server 192.168.1.1 R2(config)# ip domain name test.local R2# ping pc1